

Pharmaniaga Metoprolol Tablet

DESCRIPTION

White, round, biconvex tablet with Pharmaniaga icon on one side and '45' breakline '45' on the other.

COMPOSITION

Each tablet contains Metoprolol tartrate 100mg.

PHARMACODYNAMIC PROPERTIES

Pharmacotherapeutic group: Beta-receptor blocker, selective.

ATC code: C07A B02

Metoprolol is a beta1-selective receptor blocker, i.e., metoprolol affects the beta1-receptors of the heart in lower doses than needed to affect beta2-receptors in peripheral vessels and bronchi. At increasing doses, the beta1-selectivity may decrease.

Metoprolol has no beta-stimulating effect and has little membrane-stimulating effect. Beta receptor blockers have negative inotropic and chronotropic effect.

Metoprolol therapy reduces the effect of catecholamines in association with physical and psychic strain and gives lower heart rate, cardiac output and blood pressure. In stress situations with an increased release of adrenaline from the adrenal glands, metoprolol does not prevent the normal physiological vascular dilation. In therapeutic doses, metoprolol has less contractile effect on the bronchial muscles than non-selective beta-blockers. This property enables treatment of patients with bronchial asthma or other pronounced obstructive lung diseases with metoprolol in combination with beta2-receptor stimulants. Metoprolol influences insulin release and carbohydrate metabolism to less extent than non-selective beta-blockers and therefore it can also be given to patients with diabetes mellitus. The cardiovascular reaction in hypoglycaemia, e.g., tachycardia, is less influenced by metoprolol and the return of blood sugar level to normal is faster than for non-selective beta-receptor blockers.

In hypertension, metoprolol lowers the blood pressure significantly for more than 24 hours both in lying and standing positions as well as during exercise. In treatment with metoprolol an increase in the peripheral vascular resistance is observed initially. In long-term treatment, however, the obtained lowering in blood pressure may be due to reduced peripheral vascular resistance and unchanged cardiac output. In males with moderate/severe hypertension, metoprolol reduces the risk of cardiovascular death. There is no electrolyte imbalance.

In tachyarrhythmias the effect of increased sympatholytic activity is blocked and this gives a lower heart rate primarily by reduced automatization in the pacemaker cells, but also through a prolonged supraventricular conduction time.

Metoprolol has shown fast and effective amelioration of symptoms in thyrotoxicosis. Increased T3-values may decrease with high dose metoprolol. T4 are not affected.

Metoprolol reduces the risk of reinfarction and cardiac death, especially sudden death after myocardial infarction.

PHARMACOKINETIC PROPERTIES

The bioavailability of metoprolol is 40-50%. Maximal beta-blockade is reached after 1-2 hours. After per oral once-daily dosage of 100 mg the effect on the heart rate is still pronounced after 12 hours. Metoprolol is metabolized in the liver mainly by CYP2D6. Three main metabolites have been identified, though none has a beta-blocking effect of clinical importance. The half-life in plasma is 3-5 hours. Metoprolol is excreted to approximately 5% in unchanged form via the kidneys, the remaining dose as metabolites.

Preclinical safety data:

Metoprolol has been tested clinically to a very large extent.

INDICATIONS

- Hypertension: to reduce pressure and to reduce the risk of cardiovascular and coronary mortality (including sudden death), end morbidity.
- Angina pectoris.
- Disturbances of cardiac rhythm including especially supraventricular tachycardia.
- Confirmed or suspected myocardial infarction - prevention of cardiac death and reinfarction.
- Functional heart disorders with palpitations.
- Migraine prophylaxis.
- Hyperthyroidism.

CONTRAINDICATIONS

- Cardiogenic shock.
- Sick-sinus syndrome (provided there is no pacemaker).
- AV-block of second and third degree.
- Patients with unstable, decompensated heart failure (pulmonary oedema, hypoperfusion or hypotension), and patients with continuous or intermittent inotropic therapy acting through beta-receptor agonism.
- Symptomatic bradycardia or hypotension. Metoprolol should not be given to patients with suspected acute myocardial infarction as long as the heart rate is <45 beats/min, the P-Q interval is >0.24 sec or the systolic blood pressure is <100 mm Hg.
- Serious peripheral vascular disease with gangrene threat.
- Hypersensitivity to the active substance, to other beta-blockers or to any of the excipients.

WARNINGS AND PRECAUTIONS

Intravenous administration of verapamil should not be given to patients treated with betablockers.

Metoprolol may aggravate the symptoms of peripheral arterial circulatory disorders e.g., intermittent claudication. Severely impaired renal function. Serious acute conditions with metabolic acidosis. Combination treatment with digitalis.

Metoprolol should not be given to patients with latent or manifested heart insufficiency without concomitant treatment. In patients with Prinzmetal's angina the frequency and the extent of angina attacks may

increase due to alpha-receptor mediated contraction of the coronary vessels. For this reason, non-selective beta-blockers must not be used in these patients. Beta 1-selective receptor blockers should be used with caution.

In bronchial asthma or other chronic obstructive lung diseases, adequate bronchodilating therapy should be given concomitantly. The dose of beta2-stimulants may need to be increased.

During treatment with metoprolol the risk for interfering with carbohydrate metabolism or masking hypoglycaemia is less than with non-selective beta-blockers.

Very rarely, a pre-existing AV conduction disorder of moderate degree may become aggravated (possibly leading to AV block).

Treatment with beta-blockers may aggravate the treatment of an anaphylactic reaction. Adrenaline treatment in normal dose does not always give the expected therapeutic effect.

If metoprolol is given to a patient with pheochromocytoma, treatment with an alpha-blocker should be considered.

Withdrawal of metoprolol should, when possible, be done gradually over 2 weeks. The dose should be reduced gradually until a final dose of 25 mg is reached (half a 50 mg tablet). During this period especially patients with known ischemic heart disease should be kept under close surveillance. The risk for coronary events, including sudden death, may increase during withdrawal of beta-blockers.

Prior to surgery the anaesthetist should be informed that the patient is receiving metoprolol. It is not recommended to stop beta-blocker treatment in patients undergoing surgery. Acute initiation of high-dose metoprolol to patients undergoing non-cardiac surgery should be avoided, since it has been associated with bradycardia, hypotension and stroke including fatal outcome in patients with cardiovascular risk factors.

The tablet contains lactose. Patients with any of the following rare hereditary conditions should not use this medicine: galactose intolerance, total lactase deficiency or glucose-galactose malabsorption.

INTERACTIONS WITH OTHER MEDICAMENTS

Metoprolol is a CYP2D6-substrate. Drugs that inhibit CYP2D6 can have an effect on the plasma concentration of metoprolol. Examples of drugs that inhibit CYP2D6 are quinidine, terbinafine, paroxetine, fluoxetine, sertraline, celecoxib, propafenone and difenhydramine. When treatment with these drugs is initiated, the dose of metoprolol might have to be reduced for patients treated with metoprolol.

The following combinations with Betaloc should be avoided:

Barbituric acid derivatives: Barbiturates (investigated for pentobarbital) induce the metabolism of metoprolol by enzyme induction.

Propafenone: Upon administration of propafenone to patients on metoprolol therapy, the plasma concentrations of metoprolol increased 2-5 fold and the patients experienced side effects typical of metoprolol. The interaction is probably explained by the fact that propafenone, similarly to quinidine, inhibits the metabolism of metoprolol via cytochrome P450 2D6. The combination is probably difficult to handle since propafenone also has beta-receptor blocking properties.

Verapamil: In combination with beta-receptor blocking drugs (described for atenolol, propranolol and pindolol) verapamil may cause bradycardia and fall in blood pressure. Verapamil and beta-blockers have additive inhibitory effects on AV-conduction and sinus node function.

The following combinations with metoprolol may require modified drug dosage:

Amiodarone: Patients treated with amiodarone may develop pronounced sinus bradycardia when treated simultaneously with metoprolol. Amiodarone has extremely long half-life (around 50 days), which implies that interactions can occur for a long time after withdrawal of the drug.

Antiarrhythmics, class I: Class I-antiarrhythmics and beta-receptor blocking drugs have additive negative inotropic effects which may result in serious haemodynamic side effects in patients with impaired left ventricular function. The combination should also be avoided in "sick sinus syndrome" and pathological AV-conduction. The interaction is best documented for disopyramide.

Non-steroidal anti-inflammatory/ antirheumatic drugs: NSAID-antiphlogistics have been shown to counteract the antihypertensive effect of beta-receptor blocking drugs. Primarily, indomethacin has been studied. This interaction probably does not occur with sulindac. A negative interaction study on diclofenac has been performed.

Diphenhydramin: Diphenhydramin decreases (2.5 times) clearance of metoprolol to alphas-hydroxymetoprolol via CYP 2D6 in fast hydroxylating persons. The effects of metoprolol are enhanced. Diphenhydramin may probably inhibit the metabolism of other CYP 2D6 substrates.

Digitalis glycosides: Digitalis glycosides in association with beta-blockers, may increase atrioventricular conduction time and may induce bradycardia.

Diltiazem: Diltiazem and beta-receptor blockers have additive inhibitory effects on the AV conduction and sinus node function. Pronounced bradycardia has been observed during combination treatment with diltiazem.

Epinephrine: There are reports on patients treated with non-selective beta-receptor blockers (including pindolol and propranolol) that developed pronounced hypertension and bradycardia after

administration of epinephrine (adrenaline). These clinical observations have been confirmed in studies in healthy volunteers. It has also been suggested that epinephrine in local anaesthetics may provoke these reactions upon intravascular administration. The risk is probably less with cardioselective beta-receptor blockers.

Phenylpropanolamine: Phenylpropanolamine (norephedrine) in single doses of 50 mg may increase the diastolic blood pressure to pathological values in healthy volunteers. Propranolol generally counteracts the rise in blood pressure induced by phenylpropanolamine. However, beta-receptor blockers may provoke paradoxical hypertensive reactions in patients who take high doses of phenylpropanolamine. Hypertensive crises during treatment with only phenylpropanolamine have been described in a couple of cases.

Quinidine: Quinidine inhibits the metabolism of metoprolol in so-called rapid hydroxylators (more than 90% in Sweden) with markedly elevated plasma levels and enhanced betablockade as a result. A corresponding interaction might occur with other beta-blockers metabolized by the same enzyme (cytochrome P450 2D6).

Clonidine: The hypertensive reaction when clonidine is suddenly withdrawn may be potentiated by beta-blockers. If concomitant treatment with clonidine is to be discontinued, the beta-blocker medication should be withdrawn several days before clonidine.

Rifampicin: Rifampicin may induce the metabolism of metoprolol resulting in decreased plasma levels.

Patients receiving concomitant treatment with other beta-blockers (i.e., eye drops) or MAO inhibitors should be kept under close surveillance. In patients receiving beta-receptor blocker therapy, inhalation anaesthetics enhance the cardio-depressant effect. The dosages of oral antidiabetics may have to be readjusted in patients receiving beta-blockers. The plasma concentration of metoprolol can increase when cimetidine or hydralazine are administered simultaneously.

ADVERSE REACTIONS

Metoprolol is well tolerated, and the undesirable effects are generally mild and reversible. The most commonly reported adverse reactions during treatment is fatigue. Gangrene (in patients with severe peripheral circulatory disorder), thrombocytopenia and agranulocytosis may occur very rarely.

	Very common	Common	Uncommon	Rare	Very rare
Blood and lymphatic system disorders					Thrombocytopenia, leukopenia
Endocrine disorders				Deterioration of latent diabetes mellitus	
Metabolism and nutrition disorders			Weight gain		
Psychiatric disorders			Depression, concentration problems, drowsiness or insomnia, nightmares	Nervousness, anxiety	Forgetfulness or memory impairment, confusion, hallucination, personality

					changes (e.g. mood changes)
Nervous system disorders		Dizziness, headache	Paraesthesia		
Eye disorders				Visual disturbances, dry or irritated eyes, conjunctivitis	
Ear and labyrinth disorders					Tinnitus, hearing problems
Cardiac disorders		Bradycardia, balance disturbances (very rarely with associated syncope), palpitations	Temporary exacerbation of symptoms of heart failure, first-degree atrioventricular block, precordial pain	Functional heart symptoms, heart arrhythmia, conductivity disturbances	
Vascular disorders	Pronounced blood pressure drop and orthostatic hypotension, very rarely with syncope	Cold hands and feet			Necrosis in patients with severe peripheral vascular disorders prior to treatment, exacerbation of claudication intermittent or Raynaud's syndrome
Respiratory, thoracic and mediastinal disorders		Functional dyspnea	Bronchospasms	Rhinitis	
Gastrointestinal disorders		Nausea, abdominal pain, diarrhea, constipation	Vomiting	Dryness of mouth	Taste disturbances
Hepatobiliary disorders				Abnormal LFT values	Hepatitis
Skin and subcutaneous tissue disorders			Rash (psoriasis-like urticaria and dystrophic cutaneous lesions), increased perspiration Hypersensitivity reactions in the skin	Hair loss.	Light hypersensitivity reactions, exacerbation of psoriasis, new psoriasis manifestation, psoriasis-like dermatological changes
Musculoskeletal and connective tissue disorders			Muscle spasms		Arthralgia, muscle weakness
Reproductive system and breast disorders				Impotence and other sexual dysfunctions, induration of penis (Peyronie's syndrome)	
General disorders and administration site conditions	Fatigue		Oedema		

RECOMMENDED DOSAGE

Dosage should be adjusted individually to avoid bradycardia. The tablets should be taken on an empty stomach. Concomitant intake of food increases the bioavailability of metoprolol with 40%.

Hypertension: 100-200 mg daily divided on one or two occasions. The tablets should be taken in the morning if once-daily dosage. In patients not responding to 200 mg, the dose could be combined with other antihypertensive agents, preferably diuretics and calcium antagonists of the dihydropyridine type, or increased.

Angina pectoris: 100-200 mg daily divided on one or two occasions. If needed, the dose can be combined with nitrates or increased.

Cardiac arrhythmias: 100-200 mg daily divided on two occasions. If needed, other antiarrhythmic agents may be added.

Prophylactic therapy after myocardial infarction: As maintenance dosage, 100 mg morning and evening.

Functional heart disorders with palpitations: 100 mg once daily, given as a single dose in the morning. If needed, the dose can be increased to 200 mg.

Migraine prophylaxis: 100-200 mg daily divided on two occasions.

Hyperthyroidism: The recommended dosage is 150-200 mg daily, divided in 3-4 doses. If needed, the dose can be increased.

Impaired renal function: The elimination rate is insignificantly affected by renal function, and dose adjustment is not needed in patients with impaired renal function.

Impaired hepatic function: Usually metoprolol is given in the same dose to patients suffering from liver cirrhosis as to patients with normal liver function.

Only, when there are signs of serious impairment of liver function (e.g., shunt-operated patients) a dose reduction should be considered.

Elderly: Dose adjustment is not needed.

Children: There is limited experience with metoprolol treatment in children.

PREGNANCY AND LACTATION

Pregnancy: Metoprolol should not be given during pregnancy and lactation unless its use is considered essential. In general, beta-blockers reduce placental perfusion, which has been associated with growth retardation, intrauterine death, abortion and early labour. It is therefore suggested that appropriate maternofetal monitoring be performed in pregnant women treated with metoprolol.

Beta-receptor blockers may cause bradycardia, in the foetus and in the newborn infant. This should be considered if these drugs are prescribed in the last trimester and in association with delivery.

Metoprolol should gradually be withdrawn 48-72 hours before planned childbirth. If this is not possible the newborn infant should be supervised during 48-72 hours postpartum for signs and symptoms of beta blockade (e.g., heart and lung complications).

Lactation:

Metoprolol is concentrated in human breast milk in a quantity that corresponds to approximately three times the quantity found in the plasma of the mother. The risk for harmful reactions with respect to the breast-feeding child seems to be low at therapeutic doses of the medicine. The breast-feeding child should however be observed regarding signs of beta-blockade.

OVERDOSAGE

Toxicity: 7.5 g to an adult caused lethal intoxication. 100 mg to a 5-year old gave no symptoms after gastric lavage. 450 mg to a 12-year old and 1.4 g to an adult gave moderate intoxication, 2.5 g to an adult caused serious intoxication, and 7.5 g to an adult gave very serious intoxication.

Symptoms: Cardiovascular symptoms are most important, but in some cases, especially in children and young individuals, CNS symptoms and respiratory depression may dominate. Bradycardia, AV-block I-III, QT-prolongation (exceptional cases), asystole, fall in blood pressure, poor peripheral perfusion, cardiac insufficiency, cardiogenic shock. Respiratory depression, apnoea. Others: Fatigue, confusion, unconsciousness, fine tremor, cramps, perspiration, paraesthesiae, bronchospasm, nausea, vomiting, possibly oesophageal spasm, hypoglycaemia (especially in children) or hyperglycaemia, hyperkalaemia. Effect on the kidneys. Transient myasthenic syndrome. Concomitant ingestion of alcohol, antihypertensives, quinidine or barbiturates may aggravate the patient's condition. The first signs of overdosing may be seen 20 minutes to 2 hours after ingestion.

Management: Care should be provided at a unit that can offer suitable support measures, monitoring and supervision.

If justified, gastric lavage and/or activated charcoal can be used.

Atropine, adrenoceptor stimulant or pacemaker for treatment of bradycardia and conduction disorders.

Intubation and mechanical ventilation should be done with very broad indication. Pacemaker is option. With circulatory arrest in connection with overdose, resuscitation measures for several hours may be warranted.

Hypotension, acute myocardial infarction and shock to be treated with appropriate volume expansion, administration of glucagon (followed by intravenous infusion of glucagon if necessary), intravenous administration of an adrenoceptor stimulant, such as dobutamine, with the addition of α receptor agonists upon vasodilation. Intravenous use of Ca^{2+} may also be considered.

Bronchospasm can usually be reversed by bronchodilators.

EFFECT ON ABILITY TO DRIVE AND USE MACHINES

As dizziness and fatigue may occur in metoprolol treatment, this should be considered when strict attention is required, e.g., when driving or operating machines.

ROUTE OF ADMINISTRATION

Oral.

STORAGE CONDITIONS

Store below 30°C.
Protect from light.

SHELF LIFE

Product should not be used the expiry date imprinted on the product packaging.

PRESENTATION

In blisters of 10, 20, 30, 50, 100, 250, 500 and 1000 tablets.

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PRODUCT REGISTRATION HOLDER/ MANUFACTURER: PHARMANIAGA MANUFACTURING BERHAD (198001006232)

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